

ASAD KHAN

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SUMMARY

Backend-focused Software Engineering student with hands-on experience developing secure, scalable Java Spring Boot applications and RESTful APIs. Skilled in JWT authentication, MySQL, JPA/Hibernate, and backend system design, with experience building AI-powered applications and solving 600+ LeetCode problems. Passionate about backend engineering, distributed systems, and scalable software development.

EDUCATION

Galgotias College of Engineering and Technology

B.Tech – Information Technology | GPA: 7.52

Greater Noida, India

Expected Graduation: 2027

TECHNICAL SKILLS

Languages:	Java, Python, C++
Backend:	Spring Boot, REST APIs, Spring Security, JWT Authentication
Databases:	MySQL, JPA, Hibernate
Core CS:	Data Structures & Algorithms, Object-Oriented Programming, DBMS, Operating Systems
Developer Tools:	Git, GitHub, Maven, Postman
AI/ML:	TensorFlow, Scikit-learn, Pandas, NumPy
Cloud:	AWS
Deployment:	Vercel, Render

PROJECTS

ExoNova AI Exoplanet Discovery Platform | *Python, TensorFlow, FastAPI, React - ISRO Bharatiya Antariksh Hackathon* | [GitHub](#) *June 2026*

- Developed an AI-powered platform to assist in identifying potential exoplanet candidates using astronomical datasets from the NASA Exoplanet Archive and machine learning.
- Implemented an end-to-end pipeline for preprocessing, feature extraction, prediction, and explainable AI visualizations.
- Integrated interactive dashboards, transit signal analysis, and confidence-based predictions to support scientific exploration.
- Engineered a scalable full-stack application with secure backend APIs and an intuitive frontend for researchers and students.

Finance Dashboard | *Java, Spring Boot, REST APIs, MySQL, JWT, JPA/Hibernate* | [GitHub](#) *Jan 2026 – Mar 2026*

- Engineered 10+ REST API endpoints using Java and Spring Boot to handle financial transaction management, supporting full CRUD operations with structured error handling and HTTP status code compliance.
- Secured the application with JWT-based authentication and role-based access control via Spring Security, eliminating unauthorized access to sensitive financial data.
- Designed a normalized relational database schema in MySQL with 5+ tables and integrated JPA/Hibernate ORM to simplify database interactions and improve query maintainability.
- Optimized database access using indexing and lazy loading to improve query performance and application responsiveness.

Air Quality Prediction System | *Python, Scikit-learn, Pandas, NumPy* | [GitHub](#) *Apr 2026 – May 2026*

- Built a machine learning pipeline to predict Air Quality Index (AQI) from environmental sensor data, achieving ~85% prediction accuracy using a Random Forest classifier trained on 10,000+ data points.
- Performed end-to-end data preprocessing — missing value imputation, outlier removal, and feature engineering using Pandas and NumPy — reducing dataset noise and improving model performance by ~12%.
- Evaluated and compared 3 ML models (Logistic Regression, Decision Tree, Random Forest) using cross-validation and metrics including accuracy, precision, and recall; selected best-fit model for deployment readiness.
- Structured model outputs as clean JSON-serializable predictions, enabling seamless integration with a Spring Boot backend API.

EXPERIENCE

Technical Team Member · *GeeksforGeeks Student Chapter, GCET*

2024 – Present

- Co-organized 3+ technical events and workshops, collectively engaging 150+ student participants across campus.
- Conducted backend development sessions covering Java, Spring Boot, REST APIs, SQL, and Git for 30+ students.

- Collaborated with a 10-member technical team to organize coding events, workshops, and hackathons while improving execution timelines.

ACHIEVEMENTS & ACTIVITIES

- **600+ LeetCode problems solved** — consistent practice in arrays, graphs, dynamic programming, and trees; focused on optimizing time and space complexity.
- **ISRO Bharatiya Antariksh Hackathon 2026** — developed ExoNova, an AI-powered platform for exoplanet candidate analysis using machine learning and astronomical datasets.

CERTIFICATIONS

- Tata GenAI Powered Data Analytics Job Simulation (Forage)
- Walmart Advanced Software Engineering Job Simulation (Forage)
- AWS Certified Developer - Associate (Infosys Springboard)
- Oracle AI Foundations Associate (Oracle)
- Java (Basic) - HackerRank